

ABHYUDAYA GUPTA

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EDUCATION

The Shri Ram School Moulisari, Gurugram

2020 - Present

Grades 11 & 12 IBDP

Relevant Coursework:

- Higher Level: Math Analysis & Approaches, Physics, Chemistry, Economics
- Standard Level: English Language & Literature, Hindi

Grade 9 - A*A*A*A*A*AABC (IGCSE)

Grade 10 - A*A*AAAAAAB (IGCSE)

Grade 11 - 39/42 (IBDP)

Predicted IB: 44/45

SAT - 1500 (M: 780; EBRW: 720)

AP - Computer Science A (5); Microeconomics (5), Physics 1: Algebra-Based (4), Calculus BC (4)

USACO Platinum, Scored 1000/1000 in USACO Gold

Double AIME qualified

LEADERSHIP

Chief Developer & Experience Designer, STEMXplore

2025 - Present

Educational VR immersion simulating Physics/Chemistry/CS lab experiments for students lacking STEM lab access

- Recreated 9 lab experiments in C#, reaching 50+ schools across 11 states in India; impacting 25,000+ students in Grades 7-12
- Ranked in the top 1.1% globally (Top 250) at the Blue Ocean Competition
- Aligned the curriculum with India's National Education Policy 2020 and commended by Founder, Extramarks Education Foundation

Dept. Head, Information, Communication & Technology (ICT), TSRS Moulisari

2024 - 2026

Principal-appointed Student Council role overseeing all inter- and intra-school tech events

- Led a team of 30+ to host ShriteQ, North India's #1 tech fest, drawing 1,000+ footfall internationally
- Developed a cybersecurity module for Grades 4-12 that was integrated into the school curriculum
- Created a low-level, pointer- and bit-manipulation-based programming language compiler for the event's task force
- Progressed from Department Secretary (Grade 11) to Department Head (Grade 12); oversaw AV production for school assemblies and managed all digital infrastructure for school-wide events.

Team Head, Space Settlement Design Competition (SSDC)

2023 - 2026

International competition for Grades 9-12 to design futuristic space colonies in teams of 8-12

- Assembled and led a 12-member team to program a space colony simulation, earning Runner-up at the 2026 Asia-level competition against 30+ teams
- Directed algorithmic modeling and futuristic technology R&D, including an all-purpose quadcopter drone and a formulated, tested solid rocket fuel
- Progressed from Vice Team Leader (national level) to overseeing operations & infrastructure, to Team Head, qualifying to the Asia level in consecutive years

Designer & Creator, Mission Gaganyaan

2024 - 2025

Astrophysics-themed mystery room simulating a mission to the moon

- Designed and built an immersive physics learning experience for 400+ children at the non-profit Sapno Ke Pankh
- Awarded Innovator's Award in school for the project (design and educational impact)

RESEARCH

Published Researcher & Mentee, Dr. Kuldeep Saha

2025 - 2026

Graph-based modeling of fracture patterns in irregular mosaics using Topological Data Analysis (TDA)

- Computed Forman-Ricci curvature in Python using edge detection to derive graph representations from data and predict crack propagation

- Paper published on arXiv

Independent Researcher, Swarm Optimization

2026 - Present

- Tested a range of algorithms to determine the most optimal and efficient approach to the Traveling Salesman Problem

SUMMER PROGRAMS

Participant Scholar, University of Chicago

2026

3-week residential program in Discrete Mathematics

- Studied discrete mathematics and participated in all House Games, finishing 2nd by 4 points

Scholar, ISRO Pragyan Aerospace & Robotics Bootcamp

2025

- Trained by ISRO educators in aerospace systems; built an autonomous quadcopter drone and rover, tested solid rocket fuel, and implemented A* and RRT navigation algorithms

Fellow, Young Technology Scholars (YTS), Plaksha University

2024

2-week residential CS & engineering program with hands-on coding, hardware, and an independent project

- Built an autonomous car programmed to communicate via NRF and detect obstacles using ultrasonic sensors; secured Runner-up in a competitive race against 25+ teams under Ivy+ faculty mentorship

INTERNSHIPS

Summer Intern, Software Engineering Wing, IRAME Tech - <https://www.irame.ai/>

2025

AI-powered SaaS platform automating audits, analytics, and compliance workflows across six industries

- Built a Python prototype for AI-driven market research using REST APIs, FastAPI + Uvicorn, and web scrapers
- Analyzed competitive intelligence to inform product positioning across target industry verticals.

Co-Founder, Website Designer & Photography Head

2025

Aakash Ke Taare, a nonprofit advancing performing arts education for students from low-income families

- Designed the organization's website, helping grow new enrollments
- Led photography and theatre workshops, expanding arts access for students in rural communities

ACADEMIC HONORS & AWARDS

- USACO Platinum
- AMC 12: 124.5/150, with double qualification for AIME (score of 7)
- Blue Ocean Global Competition: Top 250, Top 1.1% Globally (International)
- CREST Gold Award
- AP Scholar Award
- Subject Excellence Award for scoring A* in 6 subjects in the Pre-Boards, Grade 10 (School)
- Triveni Award for Extracurriculars
- Innovator's Award for Mission Gaganyaan
- Srijan Award for CAS

EXTRACURRICULARS

Competitive Programmer (2024 - Present)

- Competed in multiple competitive coding competitions, including the TechVViz Hackathon, Robotronics, and the ShriteQ Hackathon

Participant, Ananta ASPEN Leadership School (2025)

- Engaged in design thinking, theatre, and leadership training

Participant, Lodha Tech for Change Competition (2025)

Advanced Bootcamp, Intel AI for Youth curriculum

- Worked with a team through the 'Demystify' and 'Inspire and Explore' modules to design an AI statistical model predicting forest fires, including a written problem statement and proposed solution